

Jay Barot

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EDUCATION

Vanderbilt University

M.S. completed, Ph.D. Computer Science, GPA 3.93/4

Nashville, USA

Aug 2022 – Present

University of Mumbai

B.E. Information Technology

Mumbai, IND

Aug 2016 – Oct 2020

EXPERIENCE

Software Developer

Syrinx

Tokyo, JPN (Remote)

Dec 2021 – Jun 2022

Empowering laryngeal cancer survivors with state-of-the-art voice-restoration technology.

- Developed and implemented advanced algorithmic optimizations, resulting in a significant reduction in signal noise by 70%, enhancing data accuracy and quality.

Lead Programmer/Analyst

Ximple Solution

Maryland, USA (Hybrid)

Jun 2019 – Apr 2021

Delivered enterprise-grade ERP and ecommerce software for distribution-focused businesses.

- Designed and built a suite of 40+ Java RESTful microservices, with a focus on reusability that cut ongoing maintenance costs.
- Built full stack applications for next-generation ERP and B2B ecommerce platforms using Angular.
- Improved UI/UX and service respond speed, lifting customer satisfaction scores significantly.
- Trained and mentored incoming engineers, improving team ramp-up time.

SELECTIVE PROJECTS

TruEye

- Developed a region-aware model for fine-grained synthetic image detection and localization that generalizes to unseen generators, achieving 93.70 IoU and 96.73 F1 on held-out generators while enabling real-time inference ($\sim 100\times$ faster than LLM-based baselines), deployed via an interactive web interface.
- Built *FineSyn*, a benchmark for cross-domain authenticity learning, via an end-to-end pipeline combining segmentation, controlled composition, diffusion/GAN synthesis, and harmonization.

Game of Papers

- Addressed the challenge of overwhelming research paper discovery by developing a BERT-powered recommendation system, improving research efficiency for users by providing context-aware, personalized suggestions.

Personal Portfolio

- Developed a sleek Angular-based portfolio showcasing research, projects, and blogs with an interactive UI.

Reverse Proxyception

- Created an ingestion service for cloud-native applications, implementing a reverse proxy mechanism to optimize request handling for scalable microservices architectures.

RESEARCH EXPERIENCE

Graduate Research Assistant | Vanderbilt University

- Developed novel datasets and scalable data pipelines, integrating data collection, annotation, preprocessing, and validation to enable new AI model capabilities.
- Synthesized findings from the literature to identify research gaps and guide end-to-end machine learning research, spanning model design, training, inference, and evaluation.
- Maintained meticulous technical documentation of datasets, experimental protocols, model architectures, and comparative analyses to ensure research reproducibility and effective collaboration.

Peer Reviewer | IEEE Transactions on Dependable and Secure Computing

- Served as a reviewer for one of the premier IEEE journals in security and dependable computing, evaluating manuscripts on topics spanning AI security, adversarial robustness, and secure systems.

PUBLICATIONS

- *ACM SACMAT* - ToneCheck: Unveiling the Impact of Dialects in Privacy Policy.
- *IJRASET* - Real Time Sentiment Analysis of Tweets on NASDAQ and NSE Stocks.
- *IJRASET* - Predicting NASDAQ and NSE Stocks using Machine Learning Algorithms.

CONFERENCE PRESENTATIONS

ACM Symposium on Access Control Models and Technologies (SACMAT) San Antonio, TX
Presented research at the UTSA School of Data Science on equitable and inclusive privacy policy design.

HONORS AND AWARDS

- **Innovation Catalyst Award** | *Vanderbilt University* Jun. 2026
Awarded for the research project “*Cost-Efficient AI for Detecting Unseen Synthetic Media.*”
- **Provisional Patent Filed** | *Advanced Synthetic Imagery Detection System* Jun. 2026
Filed a provisional patent for a novel AI-based system for detecting synthetic imagery, establishing intellectual property in the deepfake and synthetic media detection domain.
- **Fully Funded Doctoral Appointment** | *Vanderbilt University* Apr. 2024
Received a competitive five-year doctoral funding package upon transitioning to the Ph.D. program, reflecting the department’s investment in high-impact research.
- **Engineering Graduate Fellowship** | *Vanderbilt University* Aug. 2022
Awarded merit-based fellowship during M.S. studies in recognition of academic excellence and research potential in Computer Science.

CERTIFICATIONS

- **AWS** Generative AI with Large Language Models Sep. 2025

TEACHING EXPERIENCE

- **Teaching Assistant, Cyber Security (CS 4277)** | *Vanderbilt University* Spring 2026
Graded assignments and exams, helped write exam questions, and held weekly office hours for a cybersecurity course covering threat modeling, cryptography, and secure system design.
- **Teaching Assistant, Human Computer Interaction (CS 4376)** | *Vanderbilt University* Fall 2025
Delivered lectures, graded coursework, and held office hours for a course on HCI principles, usability evaluation, and user-centered design.

LEADERSHIP EXPERIENCE AND CIVIC ENGAGEMENT

Head Coordinator (Dean’s recommendation) | *Computer Society of Xaviers (CSX)*
Led 720+ students in diverse tech activities, including full stack development, computer assembly, and more.
Undergraduate Class Representative | *Class Coordination, Community Building.*

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, HTML, CSS.
Frameworks & Libraries: PyTorch, TensorFlow, Angular, React, Node.js, Flask, Django, Bootstrap.
Tools & Technologies: Docker, Kubernetes, Git, WordPress, AWS
Databases: MongoDB, PostgreSQL, MySQL.